Understanding table metrics

Files at glance

analyze_run writes four tables and a time series per run, **analyze_fraying** turns that series into fraying tables, and **analyze_replicas** pools both across replicas. Per-run files land in **<run_folder>/analysis/**, named after the trajectory. Distances are in nm unless a name ends in **_A** (Å); fractions run from 0 to 1.

File	Written by	One row per
*_nt.csv	analyze_run	Nucleotide
*_pairs.csv	analyze_run	Tracked pair: native, present in the model, or formed at least 5% of the time
*_frames.csv	analyze_run	Frame
*_summary.json	analyze_run	Run (a single record)
*_native_series.npz	analyze_run	Arrays rather than rows: H-bonds distances of the native pairs in every frame
*_fraying_pairs.csv	analyze_fraying	Native pair
*_fraying ends.csv	analyze_fraying	Helix end
*_fraying_frames.csv	analyze_fraying	Frame
*_fraying.json	analyze_fraying	Run (a single record)
<build>/replicas/pooled/<traj>_pairs.csv, _ends.csv, _summary.csv	analyze_replicas	Native pair, helix end or summary metric, across replicas
<build>/replicas/rXX/_replica.json, _done.json, _lock.json, _manifest.json	run_replicas, sync_dir	Replica bookkeeping

Changes specific to this version

The arm statistics now separate docking from end fraying, and new files cover fraying frequency and replicas.

Item	Before	Now
Arm-contact neighbour rule	Arm nucleotides within **arm_skip** of each base, on thaty base's own strand,	Arm nucleotides within **arm_skip** of either partner , on that partner's strand, are

	were ignored	ignored for both bases
occ_arm_on, **occ_arm_off**	Computed from any number of frames; one A3_P1 value rested on 3	Reported only with at least **min_contact_ps** (500ps) of contact and non-contact
Arm correlation	**arm_r_contact_occ**, over all native pairs	**arm_r_internal**, over native pairs that are not helix ends
New columns	Not reported	**end** en **contact_frames** in the per-pair table; **arm_r_internal_n** and **arm_occ_shift_n** in the summary
New files	Not reported	***_native_series.npz,** the fraying tables, the pooled replica tables and the replica bookkeeping files

The old rule let an overhang base stacked on a terminal pair count as docking.

Key terms

These are the words the tables rely on, with A3_P1 as the running example: strand A (24nt) paired with strand B (12nt) over a 12-bp design, plus a 12-overhang

Structure

Term	Meaning
Strand (chain)	One continuous nucleotide chain, given its own chain letter (A,B,...) during preparation. A new strand starts whenever the O3'-P backbone link is broken
Nucleotide label	**chain:resname resSeq**, e.g **A:DT13=** deoxythymidine 13 on chain A. **resSeq** comes from the input file, so A3_P1's strand B runs 25-36
5' and 3' ends	The two ends of a strand: the 5' end carries the first nucleotide (its phosphate is removed during preparation), the "' end the last
Duplex	Two strands paired to each other through inter-strand pairs. In A3_P1: A13-A24 with B25-B36
Stem	A run of stacked consecutive pairs (i,j), (i+1, j+1), between strands or within one. A3_P1 has a single stem
Hairpin	A stem formed by one strand folding back on itself, closed by a loop

Overhang	Unpaired nucleoteds extending beyond the end of a duplex. In A3_P1: A1-A12, a 12-nt 5’ overhang
Arm	The nucleotides outside the native inter-strand duplex (overhangs, loops unpaired end nucleotides), or the selection given with **arm=**. The arm analysis asks whether they touch duplex pairs and change their stability. In A3_P1 with the design: A1-A12
Junction pair	The duplex pair from which the arm leaves the duplex. In A3_P1: B:DA36-A:DT13
Helix end (end pair)	A native pair with no stacked native neighbour on at least one side: a duplex or stem terminus, or a pair next to a bulge or loop. The are the pairs that fray; the tables flag them **end**
Internal pair	A native pair stacked on native pairs on both sides (**end** = False)

Pairing

Term	Meaning
Design	The intended secondary structure as dot-bracket text, strands joined by **+**: **(** and **)** pair, **.** stays unpaired, **[]**, **{}** and **<>** mark pseudoknots
Built model	The solvated starting structure written by **build_system**, before minimisation. It is the reference for **in_model**, and for native pairs when no design is given.
Native pair	A pair of the target structure: in the design, or formed in the built model when no design is given.
Non-native pair	Any other complementary pair that forms during the run: misfolding, a register shift or alternative pairing.
Candidate pair	Any Watson–Crick or wobble combination on different strands, or at least 4 nt apart on one strand. Only candidates can count as formed.
Watson–Crick and wobble pairs	G·C and A·T (A·U) are Watson–Crick pairs; G·T (G·U) is the wobble pair.
Tracked H-bonds	Two per pair, purine atom first: G·C N1–N3 and O6–N4; A·T N1–N3 and N6–O4; G·T N1–O2 and O6–N3. The first one is called hb1.
Formed	Both tracked H-bond heavy-atom distances below **cut** (3.5 Å) in that frame.
Occupancy	Fraction of frames in which a pair is formed.

Opening and fraying

Term	Meaning
Breathing	A brief opening and reclosing over a few frames, not a real opening. min_open_ps and the dead band filter it out.
Opening event	A stretch during which a pair stays open. **open_events** counts those lasting at least **min_open_ps** (single cutoff); the fraying tables count closed→open transitions (dead band).
Fraying	Opening that starts at a helix end and can spread inward.
Fraying depth	In one frame, the number of consecutive open pairs counted from a helix end; 0 means the end pair is closed. Depth 2 at A3_P1's junction end = B:DA36–A:DT13 and B:DA35–A:DT14 both open.
Internal opening (bubble)	Open pairs inside a stem that no run of open pairs connect to either end.
Melted stem	Every pair of a stem open at once; for an inter-strand stem, the strands are no longer paired.
Dead band	The two-threshold rule of the fraying analysis. A pair is closed once both H-bonds are below **lo** (3.5 Å), open once one exceeds **hi** (4.5 Å), and otherwise keeps its state.
Opening rate (k_open)	Openings divided by the time spent closed: the opening rate constant of a two-state model, per μs.
Open lifetime (tau_open)	Time spent open divided by the number of closings: the mean lifetime of the open state.
Censored episode	An open or closed stretch cut off by the start or end of the run, so its true length is unknown. The rates above stay valid; **median_open_ns** leaves these out.

Arm contact

Term	Meaning

Base atoms	The nucleobase's heavy atoms: everything except the sugar (primed names) and the phosphate (P, OP1, OP2, OP3).
Arm contact (docking)	A heavy atom of an arm nucleotide within **arm_cut** (5 Å) of a base atom of either nucleotide of a pair, in one frame.
Neighbour exclusion	Arm nucleotides within **arm_skip** (2) positions of either partner, on that partner's strand, are ignored for that pair, because they are bonded to or stacked on it. For B:DA36–A:DT13, A11 and A12 are ignored.

Geometry and solvent

Term	Meaning
Frame	One saved snapshot of the trajectory; production saves one every 10 ps.
Superposition and core	A rigid fit of each frame onto a reference before measuring. The core is the set of nucleotides in native pairs, used for the fit so floppy tails don't dominate.
RMSD	Root-mean-square deviation of atom positions from a reference frame after superposition: overall drift.
RMSF	Root-mean-square fluctuation of each atom around its mean position: local flexibility.
Radius of gyration (Rg)	Root-mean-square distance of the atoms from their centre: compaction or expansion.
Sugar pucker	The shape of the deoxyribose ring: North (C3'-endo) in A-form and RNA, South (C2'-endo) in B-DNA. North shows as a positive C1'–C2'–C3'–C4' dihedral.
A-form and B-form	The two main right-handed double-helix families. B-form is DNA in water; A-form is wider and shorter, typical of RNA and dehydrated DNA.
Counterion condensation	Accumulation of cations (Na+) next to the negatively charged phosphate backbone.
Periodic image	The box is surrounded by copies of itself, so every copy of the solute is a periodic image.
Minimum image distance	The shortest distance between the solute

	and any of its images. It must stay above the 1.0 nm cutoff, or the molecule interacts with itself.
Unwrapped trajectory	Coordinates are never folded back into the box, so each strand stays in one piece. Ion–phosphate distances therefore use minimum-image distances.

Statistics and replicas

Term	Meaning
Tail	The final fraction of the analysed time (the last 50% by default), used for equilibrium averages and slopes.
Slope and plateau	The linear trend over the tail; a slope near 0 means the metric has plateaued.
Block standard error	Uncertainty of a time average from the spread of block means (5 consecutive blocks). It underestimates when slow processes are present.
Replica	An independent simulation of the same build, with its own equilibration, production and seeds.
Seed	The number that starts the random generator for initial velocities (equilibration) and thermostat noise (production). Recorded per replica and never 0, which OpenMM reads as "choose at random".
SEM	Standard error of the mean across replicas: standard deviation / √n.
Pooled rate	A rate computed from all replicas together: total openings / total closed time. It is steadier than averaging per-replica rates when events are few.

Per-nucleotide table (*_nt.csv)

Column	Information	Computation
nt	Nucleotide label	**chain:resname resSeq** from the topology, e.g. A:DT13
chain	Strand the nucleotide	Chain letter assigned during

	belongs to	preparation
resSeq	Residue number	Number carried over from the input structure, not a 0-based index
base	Nucleobase	A, C, G, T (or U)
rmsf_nm	Local flexibility: ends, loops and the arm should be high, the duplex core low	Per-atom RMS fluctuation about the mean position after superposition on the core (all heavy atoms if no core), averaged over the nucleotide's heavy atoms
paired	Whether a nucleotide is engaged in any pairing, native or not	Fraction of frames in which it is in at least one formed candidate pair
c3_endo	Local sugar pucker: near 0 (a few %) for B-DNA, near 1 for A-form	Fraction of frames with a north (C3'-endo) sugar

Per-pair table (*_pairs.csv)

Column	Information	Computation
pair	Identity of the two nucleotides	**chain:resname resSeq** of the purine, then of the pyrimidine
type	Pair chemistry	GC, AT, AU, GT or GU (purine first)
kind	Inter-strand (duplex) or intra-strand (hairpin)	Whether the two nucleotides are on different chains
native	Whether the pair belongs to the target structure	In the design, or present in the model when no design is given
end	Whether the pair sits at a helix end or next to a bulge or loop, where fraying is expected	True when the pair lacks a stacked native neighbour on at least one side; always False for non-native pairs
in_model	Whether the starting structure actually had the pair	Both H-bonds under **cut + 0.5 A** in the unrelaxed model

occupancy	Pair stability, the main per-pair figure	Fraction of frames with the pair formed (both H-bonds)
occ_hb1	Old-pipeline occupancy, for continuity	Fraction of frames with the first H-bond only under **cut**
d1_A	Typical pairing geometry; ~2.9 Å when intact	Mean first H-bond heavy-atom distance, in Å
open_events	Genuine opening events, as opposed to brief breathing	Number of stretches where the pair is open for at least **min_open_ps**
open_events_hb1	opening events with the old criterion	Same count using the first-H-bond-only definition
longest_open_ns	Whether the pair ever opened durably	Length of the longest open stretch, in ns
arm_contact	How often the arm docks on this pair	Fraction of frames where a base atom of either nucleotide is within **arm_cut** of an arm heavy atom, ignoring arm nucleotides within **arm_skip** positions of either partner on that partner's strand
contact_frames	How much contact the arm statistics rest on	Number of frames with arm contact
occ_arm_on	Stability while the arm touches the pair	Occupancy over frames with arm contact; empty when contact totals less than **min_contact_ps**
occ_arm_off	Stability while the arm is away	Occupancy over frames without contact; empty when those total less than **min_contact_ps**. Compare with **occ_arm_on** for a direct, per-pair test of arm effects

Per-frame table (*_frames.csv)

Column	Information	Computation

time_ns	Simulation time	Frame index × time step read from the trajectory (or **10 ps * stride** if the file has no usable time stamps)
q_native	Overall fraction of the designed structure that is intact	Fraction of native pairs formed in the frame
q_inter	Integrity of the inter-strand duplex(es), i.e. strand association	Same, restricted to native pairs between different strands
q_intra	Integrity of hairpins and other intra-strand structure	Same, restricted to native pairs within one strand
q_native_hb1	Old-pipeline equivalent of q_native, for comparison with earlier runs	Fraction of native pairs whose first H-bond (N1–N3 for WC, N1–O2 for wobble) is under **cut**
nonnative_pairs	Misfolding or alternative pairing	Number of formed complementary pairs that are not native (any WC/wobble combination at least 4 nt apart on the same strand, or on different strands)
rmsd_nm	Global structural drift	Heavy-atom RMSD of all nucleotides after optimal superposition on **frame 0** of the trajectory (not the built model)
rmsd_core_nm	Drift of the paired core only, insensitive to floppy tails and loops	Same as **rmsd_nm**, using only nucleotides that take part in native pairs
rg_nm	Compaction or expansion of the whole assembly	Radius of gyration of nucleotide heavy atoms, unweighted by mass
a_form	Helix family: B-DNA gives ~0, A-form gives ~1; a rising value signals an A-like transition or dehydration artefact	Fraction of nucleotides with a north (C3'-endo) sugar, detected as a positive C1'–C2'–C3'–C4' dihedral
cations_5A	Counter-ion condensation on the backbone	Number of cation atoms (Na, K, Li, Rb, Cs, Mg, Ca) within 5 Å of any phosphorus, using minimum-image periodic distances

e2e_X_nm	Extension or bending of strand X	Distance from the 5'-terminal O5' to the 3'-terminal O3' of the strand
com_XY_nm	Whether two strands stay together; a jump signals dissociation	Distance between the geometric centres of the heavy atoms of strands X and Y
min_image_nm	Whether the box is large enough: the solute must not see its own periodic copy	Closest heavy-atom distance between the solute and its 26 periodic images, every **image_every** frames (empty in between)
arm_contacts	How much of the duplex the arm is touching	Number of native pairs with arm contact in the frame, using the same per-pair neighbour rule as **arm_contact**

Run summary table (*_summary.json)

Column	Information	Computation
run, length_ns	Which trajectory, and how much of it was analysed	Folder/trajectory name (plus **max_ns** if set); time of the last frame
strands, nucleotides	System size	Counts from the topology
native_pairs, native_inter	Size of the target structure	Number of native pairs, and how many are inter-strand
native_missing_in_model	Mismatch between design and built model; non-zero means the starting structure was already off-target	Native pairs not formed in the model
native_below_50pct	Number of weak spots in the structure	Native pairs with occupancy under 0.5
min_image_nm	Worst-case periodic contact over the run	Minimum of **min_image_nm** over the sampled frames
<metric>_tail	Equilibrium value of every per-frame metric	Mean over the last **tail** fraction (50% by default) of the analysed time
<metric>_slope_per_ns	Whether the run has converged; values near 0 mean a plateau, a clear	Linear-fit slope over the same tail, for **q_native, q_inter, q_intra, a_form,**

	trend means it is still drifting	**cations_5A** and **rmsd_core_nm**
arm_nt	Size of the arm	Number of arm nucleotides
arm_contact_mean	Average docking of the arm on the duplex	Mean **arm_contact** over native pairs
arm_r_internal	Across-pair association between docking and stability; few points, so suggestive only	Pearson r between arm_contact and occupancy over internal native pairs; empty with fewer than 3 pairs or no variation
arm_r_internal_n	How many pairs the correlation rests on	Number of pairs used for **arm_r_internal**
arm_occ_shift*	Within-pair effect of the arm; negative means pairs are formed less often while the arm touches them	Mean of **occ_arm_on** − **occ_arm_off** over native pairs where both are defined
arm_occ_shift_n	How many pairs the shift rests on	Number of pairs averaged

***arm_occ_shift** shows an association, not a cause: at a junction pair, the overhang base stacking and the pair closing move together.

Native-pair time series (*_native_series.npz)

This NumPy archive holds the native pairs' H-bond distances in every frame; **analyze_run** writes it and **analyze_fraying** reads it.

Array	Content
time_ns	Time of each frame, in ns
d_nm	Frames × native pairs × 2: the two tracked H-bond heavy-atom distances of each pair, in nm
label	Pair labels, in the same format as **pair** in ***_pairs.csv**
res	Topology indices (0-based) of the two nucleotides of each pair, smaller first
res_chain	Chain index of every residue, used to find stacked neighbours
end	True for helix-end pairs

cut	H-bond threshold used by **analyze_run**, in nm

Fraying tables (*_fraying_*)

Per pair (*_fraying_pairs.csv)

Column	Information	Computation
pair	Identity of the pair	Same label as in *_pairs.csv
end	Helix-end pair	Same flag as in *_pairs.csv
p_open	How much of the time the pair is open	Fraction of frames open under the dead band
p_open_se	Rough uncertainty of **p_open** within this run	Standard error over n_blocks consecutive blocks; prefer the spread across replicas
openings, closings	How many times the pair opened, and closed again	Number of closed→open and open→closed transitions
t_closed_ns, t_open_ns	Time spent closed, and open	Frames in each state × frame interval, in ns
openings_per_100ns	Fraying frequency over the whole run	**openings** / analysed time × 100 ns
long_openings	Openings that last, comparable with open_events	Number of open episodes lasting at least **long_ps**
k_open_per_us	Opening rate	**openings** / **t_closed_ns**, expressed per μs
tau_open_ns	Mean open lifetime	**t_open_ns** / **closings**
median_open_ns	Typical open episode	Median length of open episodes seen from start to finish (censored ones left out)
max_open_ns	Longest opening	Longest open episode, in ns

Per helix end (*_fraying_ends.csv)

Column	Information	Computation
end	Which end	Label of the stem's terminal pair at that end
stem	Which stem the end belongs to	Labels of the stem's first and last pairs
p_frayed	How often this end is frayed	Fraction of frames with depth ≥ 1 (end pair open)
p_depth_ge2, p_depth_ge3	How far fraying spreads inward	Fraction of frames with at least 2, or 3, consecutive open pairs from this end
mean_depth_frayed	Typical depth once frayed	Mean depth over frames with depth ≥ 1
max_depth	Deepest fraying seen	Maximum depth over the run
frays_per_100ns	Fraying frequency of this end	Number of depth 0 → ≥ 1 transitions per 100 ns
p_internal_open	Openings inside the stem, away from the ends; expected near 0	Fraction of frames with at least one internal open pair (same value on both rows of a stem)
p_melted	Complete opening of the stem	Fraction of frames with every pair of the stem open (same value on both rows)

Per frame (*_fraying_frames.csv)

Column	Information	Computation
time_ns	Simulation time	As in ***_frames.csv**
depth[<end pair>]	Fraying depth at that end, for plotting	Consecutive open pairs counted from that end
internal[<stem>]	Internal openings in that stem	Open pairs not connected to an end

Fraying summary (*_fraying.json)

Field	Information	Computation

run	Which trajectory	Folder/trajectory name
lo_nm, hi_nm, full_open	Dead-band settings used	As passed to **analyze_fraying**
length_ns	Analysed time	Frames * frame interval
openings_per_100ns_ends	Total fraying at helix ends	Sum of **openings_per_100ns** over end pairs
openings_per_100ns_internal	Total opening of internal pairs; expected near 0	Same sum over internal pairs
p_internal_open_max	Worst internal opening	Maximum **p_internal_open** over stems
p_melted_max	Closest approach to full melting	Maximum **p_melted** over stems

Pooled replica tables (<build>/replicas/pooled/)

analyze_replicas runs both analyses on every finished replica and writes the spread across them. Rows are matched by pair label, which the shared build keeps identical across replicas.

Field and column	Information	Computation
<traj>_pairs.csv: <metric>_mean	Replica average of a per-pair metric	Mean across replicas of **occupancy**, **p_open**, **openings_per_100ns**, **long_openings**, **k_open_per_us**, **tau_open_ns**, **max_open_ns**, **arm_contact**, **occ_arm_on** and **occ_arm_off**
<traj>_pairs.csv: <metric>_sem	Uncertainty of that average	Standard deviation across replicas / √n
<traj>_pairs.csv: k_open_pooled_per_us	Opening rate from all replicas together	Total openings / total closed time, per µs
<traj>_pairs.csv: tau_open_pooled_ns	Open lifetime from all replicas together	Total open time / total closings
<traj>_ends.csv: <metric>_mean, <metric>_sem	Replica average and uncertainty per helix end	Same, for every column of ***_fraying_ends.csv**
<traj>_summary.csv: mean, sem, min, max, n	Spread of each run-level metric	One row per numeric field of ***_summary.json** and ***_fraying.json** (the latter prefixed

		fray_); **n** = replicas with a value

For **min_image_nm**, read **min**: the worst replica is the one that matters.

Replica bookkeeping files (<build>/replicas/rXX/)

These small files make replicas resumable and traceable. The tables never read them, but deleting one changes what **run_replicas** does next.

File	Information	Content
_replica.json	Provenance of the replica	Replica number, build (hash of **system.xml**), source folder, **ns**, **temp**, **dt_fs**, **save_ps**, **seed_equil**, **seed_prod**, equilibration ladder and OpenMM version
_done.json	Marks a finished replica	Session and time of completion; finished replicas are skipped, and they are the only ones pooled
_lock.json	Which session is running the replica	Session id and expiry time; other sessions skip the replica until then. Delete it to take over from a session that died
_manifest.json	File sizes at the last mirror to Drive	Written by **sync_dir**, checked by **restore_from_drive**